Critical Thinking Module 1: Linux Shell Commands

Alexander Ricciardi

Colorado State University Global

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Professor: Dr. Joseph Issa

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Computers are built from several major structural elements, especially the processor, main memory, input/output modules, and long-term storage. Operating systems are responsible for coordinating these resources so programs can run, data can move into memory, devices can exchange information, and files can be stored correctly (Bazuku et al., 2023). This hardware inspection is not only about naming parts, but also about showing how the operating system sees those parts and why their specifications matter. This paper uses Linux shell commands to examine a computer desktop system running Ubuntu 24.04.4 LTS by identifying the processor, main memory, I/O modules, and storage devices visible to the operating system. The examination results show that the system has a 13th Generation Intel Core i9-13900KS processor, 125 GiB of RAM, several PCI and USB devices, and two physical NVMe solid-state drives.

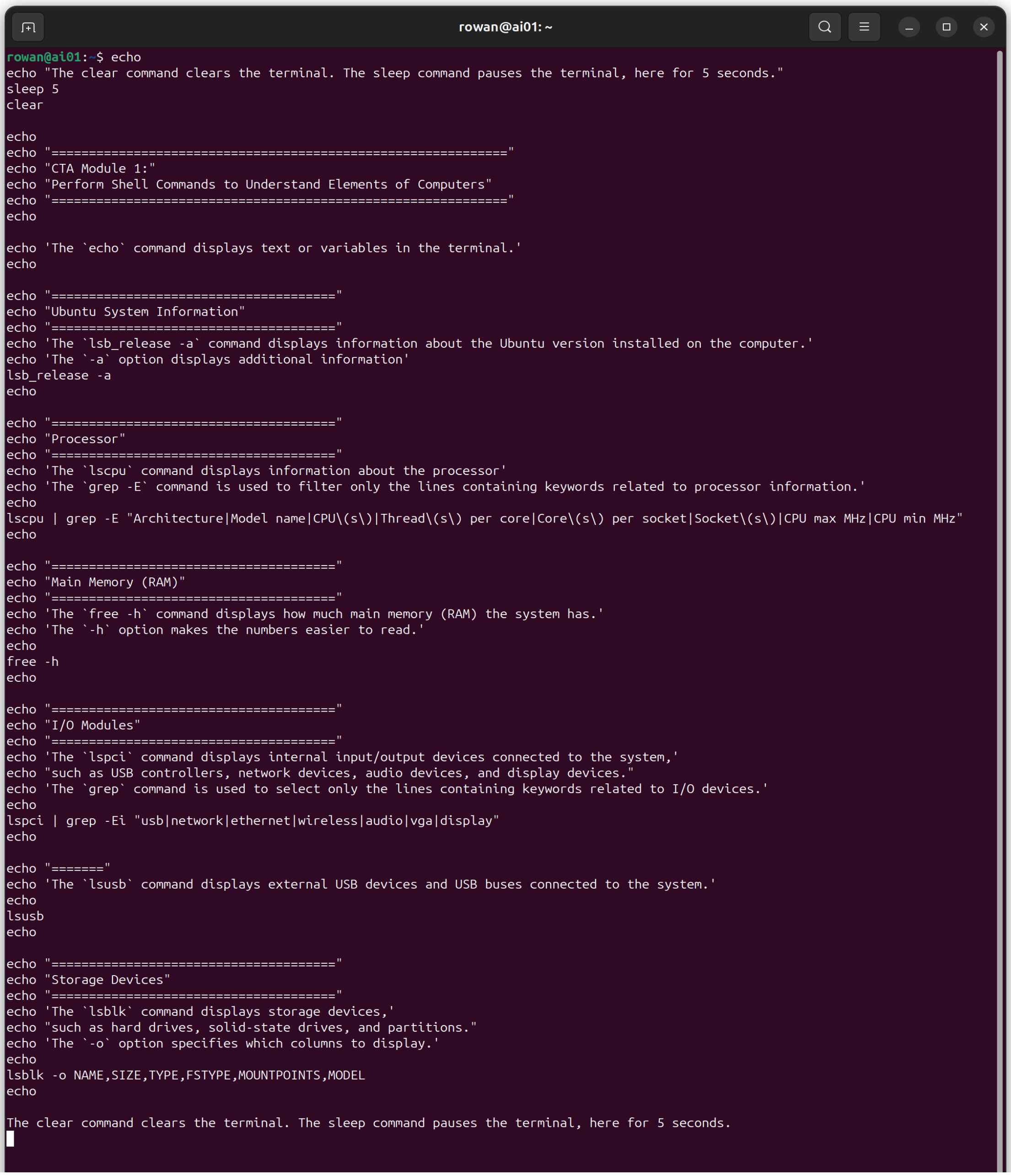
Shell Commands Used

A shell script was used to write the command, see Figure 3; however, the commands were copied and pasted to the terminal, as using a script was out of the scope of this critical thinking assignment. The commands used were`echo`, `sleep`, and `clear` to format the terminal session, see Figure 1, but the main inspection relied on `lsb_release -a`, `lscpu | grep -E ...`, `free -h`, `lspci | grep -Ei ...`, `lsusb`, and `lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINTS,MODEL`. The command references explain that `lsb_release` prints Linux distribution information, `lscpu` reports CPU architecture data, `free` reports memory usage, `lspci` lists PCI devices, `lsusb` lists USB devices, and `lsblk` lists block devices (Barabucci, 2025; Kerrisk, n.d.-a, n.d.-b, n.d.-c, n.d.-d, n.d.-e). `grep` was added to the `lscpu` and `lspci` commands so the output would stay focused on the most relevant lines.

The `lsb_release -a` output also confirmed that the system was running Ubuntu 24.04.4 LTS with the codename noble, which established the environment for the remaining results.

Figure 1

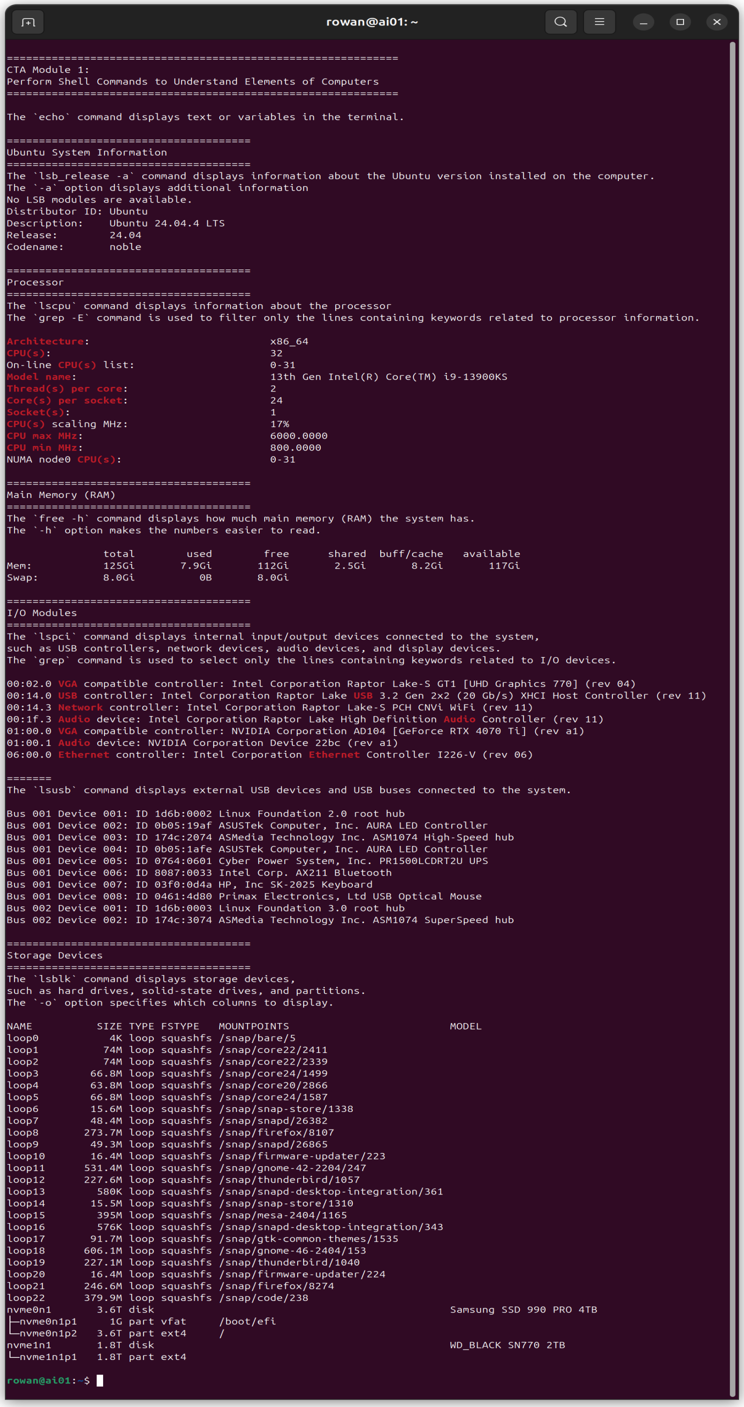
Terminal command sequence paused by `sleep 5`



Note: Figure 1 shows the command sequence while the script is paused by `sleep 5`.

Figure 2

Terminal results from the Linux hardware inspection

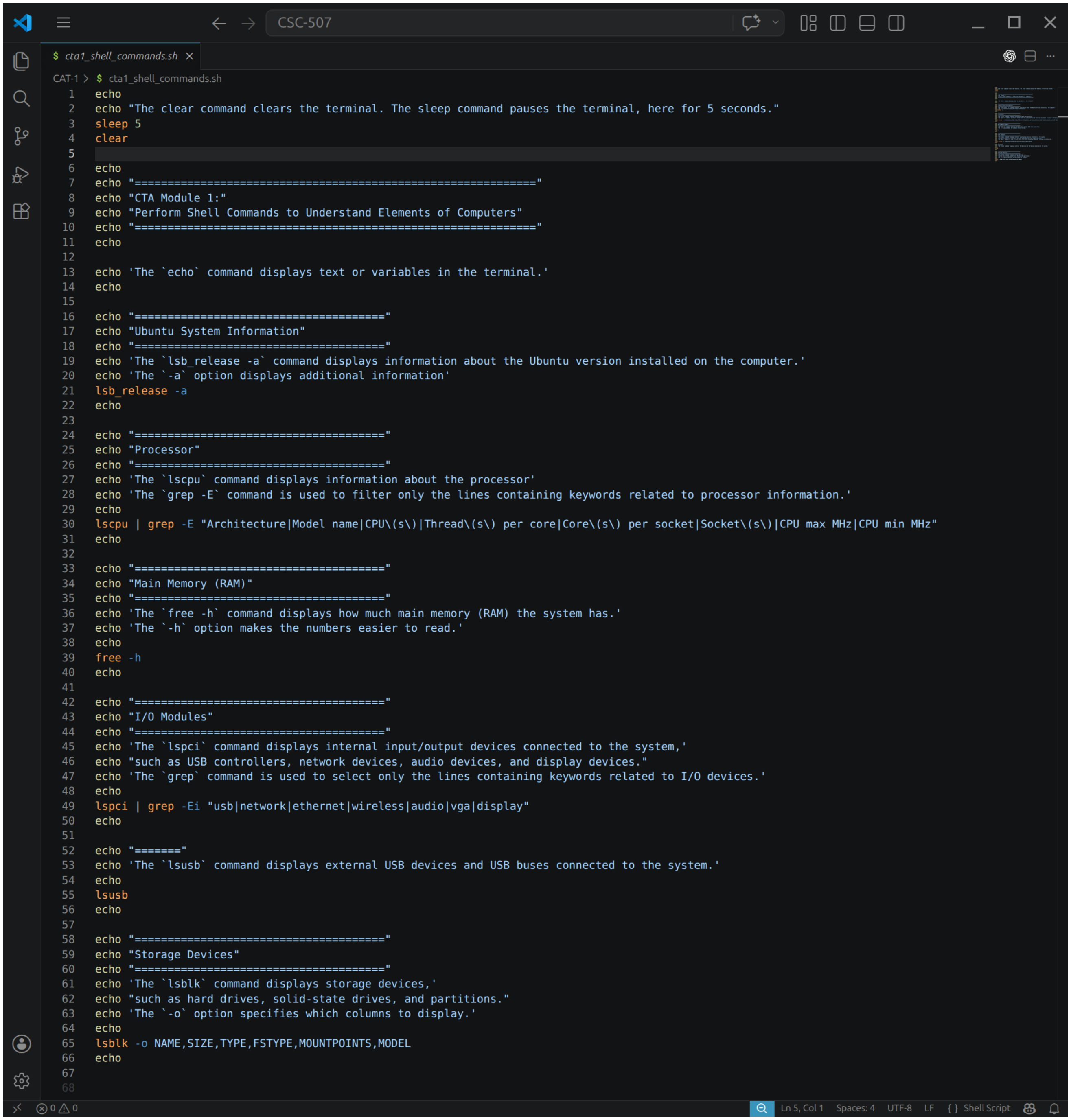


```
rowan@ai01: ~

============================================================
CTA Module 1:
Perform Shell Commands to Understand Elements of Computers
============================================================

The `echo` command displays text or variables in the terminal.

======================================
Ubuntu System Information
======================================
The `lsb_release -a` command displays information about the Ubuntu version installed on the computer.
The `-a` option displays additional information
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.4 LTS
Release:        24.04
Codename:       noble

======================================
Processor
======================================
The `lscpu` command displays information about the processor
The `grep -E` command is used to filter only the lines containing keywords related to processor information.

Architecture:                         x86_64
CPU(s):                               32
On-line CPU(s) list:                  0-31
Model name:                           13th Gen Intel(R) Core(TM) i9-13900KS
Thread(s) per core:                   2
Core(s) per socket:                   24
Socket(s):                            1
CPU(s) scaling MHz:                   17%
CPU max MHz:                          6000.0000
CPU min MHz:                          800.0000
NUMA node0 CPU(s):                    0-31

======================================
Main Memory (RAM)
======================================
The `free -h` command displays how much main memory (RAM) the system has.
The `-h` option makes the numbers easier to read.

               total        used        free      shared  buff/cache   available
Mem:           125Gi       7.9Gi       112Gi       2.5Gi       8.2Gi       117Gi
Swap:          8.0Gi          0B       8.0Gi

======================================
I/O Modules
======================================
The `lspci` command displays internal input/output devices connected to the system,
such as USB controllers, network devices, audio devices, and display devices.
The `grep` command is used to select only the lines containing keywords related to I/O devices.

00:02.0 VGA compatible controller: Intel Corporation Raptor Lake-S GT1 [UHD Graphics 770] (rev 04)
00:14.0 USB controller: Intel Corporation Raptor Lake USB 3.2 Gen 2x2 (20 Gb/s) XHCI Host Controller (rev 11)
00:14.3 Network controller: Intel Corporation Raptor Lake-S PCH CNVi WiFi (rev 11)
00:1f.3 Audio device: Intel Corporation Raptor Lake High Definition Audio Controller (rev 11)
01:00.0 VGA compatible controller: NVIDIA Corporation AD104 [GeForce RTX 4070 Ti] (rev a1)
01:00.1 Audio device: NVIDIA Corporation Device 22bc (rev a1)
06:00.0 Ethernet controller: Intel Corporation Ethernet Controller I226-V (rev 06)

=======
The `lsusb` command displays external USB devices and USB buses connected to the system.

Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 001 Device 002: ID 0b05:19af ASUSTek Computer, Inc. AURA LED Controller
Bus 001 Device 003: ID 174c:2074 ASMedia Technology Inc. ASM1074 High-Speed hub
Bus 001 Device 004: ID 0b05:1afe ASUSTek Computer, Inc. AURA LED Controller
Bus 001 Device 005: ID 0764:0601 Cyber Power System, Inc. PR1500LCDRT2U UPS
Bus 001 Device 006: ID 8087:0033 Intel Corp. AX211 Bluetooth
Bus 001 Device 007: ID 03f0:0d4a HP, Inc SK-2025 Keyboard
Bus 001 Device 008: ID 0461:4d80 Primax Electronics, Ltd USB Optical Mouse
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 002 Device 002: ID 174c:3074 ASMedia Technology Inc. ASM1074 SuperSpeed hub

======================================
Storage Devices
======================================
The `lsblk` command displays storage devices,
such as hard drives, solid-state drives, and partitions.
The `-o` option specifies which columns to display.

NAME          SIZE TYPE FSTYPE   MOUNTPOINTS                        MODEL
loop0           4K loop squashfs /snap/bare/5
loop1          74M loop squashfs /snap/core22/2411
loop2          74M loop squashfs /snap/core22/2339
loop3        66.8M loop squashfs /snap/core24/1499
loop4        63.8M loop squashfs /snap/core20/2866
loop5        66.8M loop squashfs /snap/core24/1587
loop6        15.6M loop squashfs /snap/snap-store/1338
loop7        48.4M loop squashfs /snap/snapd/26382
loop8       273.7M loop squashfs /snap/firefox/8107
loop9        49.3M loop squashfs /snap/snapd/26865
loop10       16.4M loop squashfs /snap/firmware-updater/223
loop11      531.4M loop squashfs /snap/gnome-42-2204/247
loop12      227.6M loop squashfs /snap/thunderbird/1057
loop13        580K loop squashfs /snap/snapd-desktop-integration/361
loop14       15.5M loop squashfs /snap/snap-store/1310
loop15        395M loop squashfs /snap/mesa-2404/1165
loop16        576K loop squashfs /snap/snapd-desktop-integration/343
loop17       91.7M loop squashfs /snap/gtk-common-themes/1535
loop18      606.1M loop squashfs /snap/gnome-46-2404/153
loop19      227.1M loop squashfs /snap/thunderbird/1040
loop20       16.4M loop squashfs /snap/firmware-updater/224
loop21      246.6M loop squashfs /snap/firefox/8274
loop22      379.9M loop squashfs /snap/code/238
nvme0n1       3.6T disk                                             Samsung SSD 990 PRO 4TB
├─nvme0n1p1     1G part vfat     /boot/efi
└─nvme0n1p2   3.6T part ext4     /
nvme1n1       1.8T disk                                             WD_BLACK SN770 2TB
└─nvme1n1p1   1.8T part ext4

rowan@ai01:~$
```

Note: Figure 2 shows the hardware output used for the processor, memory, I/O, and storage analysis.

Figure 3

Visual Studio Code view of the shell script



```
echo
echo "The clear command clears the terminal. The sleep command pauses the terminal, here for 5 seconds."
sleep 5
clear

echo
echo "============================================================="
echo "CTA Module 1:"
echo "Perform Shell Commands to Understand Elements of Computers"
echo "============================================================="
echo

echo 'The `echo` command displays text or variables in the terminal.'
echo

echo "======================================="
echo "Ubuntu System Information"
echo "======================================="
echo 'The `lsb_release -a` command displays information about the Ubuntu version installed on the computer.'
echo 'The `-a` option displays additional information'
lsb_release -a
echo

echo "======================================="
echo "Processor"
echo "======================================="
echo 'The `lscpu` command displays information about the processor'
echo 'The `grep -E` command is used to filter only the lines containing keywords related to processor information.'
echo
lscpu | grep -E "Architecture|Model name|CPU\(s\)|Thread\(s\) per core|Core\(s\) per socket|Socket\(s\)|CPU max MHz|CPU min MHz"
echo

echo "======================================="
echo "Main Memory (RAM)"
echo "======================================="
echo 'The `free -h` command displays how much main memory (RAM) the system has.'
echo 'The `-h` option makes the numbers easier to read.'
echo
free -h
echo

echo "======================================="
echo "I/O Modules"
echo "======================================="
echo 'The `lspci` command displays internal input/output devices connected to the system,'
echo "such as USB controllers, network devices, audio devices, and display devices."
echo 'The `grep` command is used to select only the lines containing keywords related to I/O devices.'
echo
lspci | grep -Ei "usb|network|ethernet|wireless|audio|vga|display"
echo

echo "======="
echo 'The `lsusb` command displays external USB devices and USB buses connected to the system.'
echo
lsusb
echo

echo "======================================="
echo "Storage Devices"
echo "======================================="
echo 'The `lsblk` command displays storage devices,'
echo "such as hard drives, solid-state drives, and partitions."
echo 'The `-o` option specifies which columns to display.'
echo
lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINTS,MODEL
echo
```

Note: Figure 3 shows the script-writing workflow before the commands were copied into the Linux terminal.

Processor

A processor is the main execution component of the computer. It carries out instructions and provides the computing power that the operating system must divide across running processes. In this inspection the processor was examined with `lscpu`, and the filtered output, see Figure 2, identified the architecture as x86_64 and the CPU model as a 13th Gen Intel(R) Core(TM) i9-13900KS. The same output also reported 32 CPUs visible to Linux, 24 cores per socket, 1 socket, and a frequency range from 800.0000 MHz to 6000.0000 MHz. In other words, the system was using a modern high-performance processor with substantial parallel processing capacity. From an operating-system perspective, that matters because more logical CPUs give the scheduler more execution resources to distribute across running tasks.

Main Memory

Main memory stores the data and instructions that active programs need while they are running. Bazuku et al. (2023) emphasize that operating systems depend heavily on memory coordination in order to manage program execution effectively. In this inspection, see Figure 2, main memory was examined with `free -h`, which reported 125 GiB of total RAM and 8.0 GiB of swap space. The exact “used,” “free,” and “available” values shown by `free -h` are only a snapshot, but the main structural conclusion is still clear. The machine had a very large amount of installed memory, and most of it was still available at the time of capture. In other words, the system was not under visible memory pressure. These results show why `free -h` is useful in an operating-systems context: it reports both total capacity and whether the system is close to relying heavily on swap.

I/O Modules

I/O modules move data between the computer and its external environment. The results show that Linux exposes I/O hardware through more than one bus, which is why this section relied on both `lspci` and `lsusb`. The `lspci | grep -Ei "usb|network|ethernet|wireless|audio|vga|display"` output showed, see Figure 2, major PCI-connected controllers, including Intel UHD Graphics 770, an NVIDIA GeForce RTX 4070 Ti, an Intel USB 3.2 controller, Intel Wi-Fi, Intel Ethernet Controller I226-V, and audio devices associated with both the Intel chipset and the NVIDIA graphics hardware. In other words, the PCI listing showed the internal controllers that manage display, network communication, USB support, and sound. The `lsusb` output added USB-visible devices and hubs, including Linux root hubs, ASUSTek AURA controllers, ASMedia hubs, a Cyber Power UPS, Intel AX211 Bluetooth, an HP keyboard, and a Primax USB optical mouse. These results show that the system's I/O environment included both internal controllers and external or bus-level devices. From an operating-system perspective, that matters because the OS must coordinate graphics, audio, Ethernet, wireless networking, Bluetooth, keyboard input, mouse input, USB hubs, and power-related devices at the same time.

Storage Devices

Storage devices provide long-term data retention for the operating system, installed programs, and user files. In this inspection, storage devices were examined with `lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINTS,MODEL`, which lists block devices together with size, filesystem type, mount points, and model name. That command was a strong choice because it combines hardware identity with information about how the operating system is using each device. The most important result is that the output identified two physical NVMe solid-state drives. The first was, see Figure 2, a Samsung SSD 990 PRO 4TB listed as nvme0n1 with a size of about 3.6T. The second was a WD_BLACK SN770 2TB listed as nvme1n1 with a size of about 1.8T. The Samsung drive contained the vfat EFI partition and the ext4 partition mounted at /, which means it was the active system drive. The WD_BLACK drive also had an ext4 partition, showing that it had been formatted for additional storage use.

It is important to note that the `lsblk` output also showed many loop devices. At first glance, those entries might look like extra storage drives, but that interpretation would be incorrect. The mount points under /snap/... show that these were mounted Snap package images rather than separate physical disks. In other words, the output distinguishes between real hardware storage and software-mounted block images. The system had two real internal NVMe SSDs, while the loop devices were operating-system-managed software images.

Conclusion

This hardware inspection shows how Linux command-line tools can be used to connect operating-systems concepts directly to real computer hardware. `lscpu` showed that the inspected system used a 13th Gen Intel Core i9-13900KS with 32 logical CPUs visible to Linux. `free -h` showed 125 GiB of RAM and 8.0 GiB of swap. `lspci` and `lsusb` revealed a broad I/O environment that included graphics, networking, audio, Bluetooth, USB, a keyboard, a mouse, and a UPS. `lsblk` showed two physical NVMe SSDs and also made it possible to distinguish real storage hardware from Snap-backed loop devices. Taken together, these results show that Linux does not hide the structure of the machine from the user. Instead, it provides direct tools that make the processor, memory, I/O modules, and storage devices visible in a way that supports both technical inspection and operating-systems analysis. The inspected computer appears to be a powerful modern system with substantial compute, memory, device, and storage resources.

References

Barabucci, G. (2025). *`lsb_release` - print distribution-specific information (minimal implementation)*. Ubuntu Manpage Repository. https://manpages.ubuntu.com/manpages/noble/man1/lsb_release.1.html

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